

## Homework 2: Newtonian Potential and Gravity Fields

This homework covers material from **Module 2: Newtonian Potential and Gravity Fields** and aligns with learning outcomes **LO2**.

**Related Resources:** - Lectures: M2 lecture notes (Gravitational Potential and Applications)  
- Lab: Lab 2 (Gravity of Simple Bodies) — *Strongly recommended to complete before/during HW* - Textbook: Blakely Ch. 3, 7

### Problems

1. **Problem 2.1** (Blakely Ch. 3, Problem 1) — *Foundation/Easy*

- Derive the gravitational potential and vertical attraction of a point mass. Include the mathematical expressions and units.
- **Cross-reference:** Lab 2, Part 1; M0/lecture3\_notes.qmd (vector calculus review)
- **Difficulty:** Easy (fundamental derivation using Coulomb-Newton analogy)
- **Tip:** Start with Newton's law of universal gravitation; integrate to get potential; differentiate to get field

2. **Problem 2.2** (Blakely Ch. 3, Problem 4) — *Conceptual/Moderate*

- Apply Newton's shell theorem to explain why the gravitational field inside a uniform spherical shell is zero. Extend this to the PREM model.
- **Cross-reference:** Lab 2, Part 2; M0/lecture3\_notes.qmd (coordinate systems)
- **Difficulty:** Moderate (proof technique; extension to real Earth models)
- **Tip:** Use symmetry argument; integrate over shell; relate each shell in PREM to enclosed mass

3. **Problem 2.3** (Blakely Ch. 7, Problem 2) — *Computational/Moderate*

- Calculate the free-air, Bouguer, and isostatic corrections for a gravity station at elevation 1000 m above sea level, with Bouguer plate thickness 30 m. Use standard densities ( $\rho_{\text{crust}}=2670 \text{ kg/m}^3$ ,  $\rho_{\text{water}}=1000 \text{ kg/m}^3$ ,  $\rho_{\text{mantle}}=3300 \text{ kg/m}^3$ ).
- **Cross-reference:** Lab 2, Part 3 (corrections application); M2 lecture notes
- **Difficulty:** Moderate (straightforward calculations; understanding physical meaning)
- **Tip:** Research standard formulas; check units carefully; explain physical significance of each correction

4. **Problem 2.4** (Blakely Ch. 3, Problem 6) — *Analytical/Moderate*

- Compute the gravitational potential of a solid sphere of uniform density  $\rho$  and radius  $R$ . Compare with the point mass approximation for points outside the sphere.
- **Cross-reference:** Lab 2, Part 1; M1/lecture1\_notes.qmd (spherical coordinates)

- **Difficulty:** Moderate (integration in spherical coordinates; asymptotic analysis)
- **Tip:** Integrate shell by shell; use shell theorem result; check agreement with point mass for  $r \gg R$

5. **Problem 2.5** (Blakely Ch. 3, Problem 8) — *Application/Challenging*

- Explain Gauss's law for gravity and derive the expression for excess mass from the gravity anomaly of a buried sphere.
- **Cross-reference:** Lab 2, Part 2; M1/lecture2\_notes.qmd (Green's identities); Final Project (inverse modeling)
- **Difficulty:** Challenging (synthesis of Gauss's law, BVP theory, and geophysical interpretation)
- **Tip:** Start with divergence theorem; relate field to enclosed mass; work backward from anomaly to infer source

## Submission Guidelines

- Submit solutions as a PDF or Quarto document
- Include derivations, calculations, and explanations with proper units
- Show all work step-by-step
- Include graphs/sketches where helpful (e.g., gravity vs depth plots)
- Cite references appropriately
- Expected length: 4–6 pages